

# Tree width and Grid Minors

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# Graph Operations

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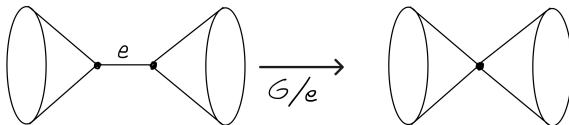
## Vertex Deletion

$$G \xrightarrow{\text{delete } v} G - v$$

## Edge Deletion

$$G \xrightarrow{\text{delete } e} G - e$$

## Edge Contraction



# Graph Minors

## Definition

A graph  $H$  is a **minor** of  $G$ ,  $H \leq G$ , if  $H$  is obtained from  $G$  by a sequence of vertex deletions, edge deletions, and edge contractions.

Add example

# Topological Minors

## Notable Results

### Theorem (Kuratowski's Theorem)

*A graph  $G$  is planar if and only if  $G$  does not have a  $K_5$  or  $K_{3,3}$  minor.*

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Equivalently, the class of graphs determined by the **forbidden minors**  $\{K_5, K_{3,3}\}$  is precisely all planar graphs.

# Notable Results

## Theorem (Robertson–Seymour Theorem)

*Every minor-closed class of graphs has a finite set of minimal forbidden minors.*

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### Theorem (Excluded Grid Theorem (Robertson and Seymour))

*Let  $r, m > 0$  be integers, and let  $G$  be a graph of treewidth at least  $r^{4m^2(r+2)}$ . Then  $G$  contains either  $K_m$  or the  $r \times r$  grid as a minor.*

## Notable Results

and many more involving surface embeddings, other graph parameters (pathwidth, ...), graph decompositions (Graph Minor Structure Theorem), etc

## Our Goal

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We will consider a proof by Diestel, Gorbunov, Jensen, Thomassen 99'

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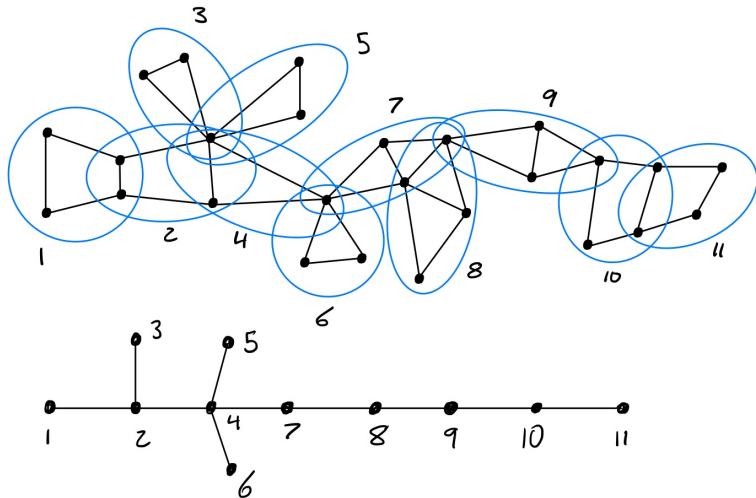
# Tree Decompositions

A collection of **bags**  $\mathcal{B} = (B_x \subseteq V(G) \mid x \in V(T))$  indexed by a tree  $T$  such that:

- 1 For every edge  $e = uv$  in  $G$  there exists a bag  $B_x$  such that  $u, v \in B_x$ .
- 2 For every vertex  $v \in V(G)$  the set  $\{x \in V(T) \mid v \in B_x\}$  induces a non-empty connected subtree of  $T$ .

Then  $(\mathcal{B}, T)$  is a **tree decomposition** of  $G$ .

# Treewidth





# TCS Applications

Many *NP – Hard* problems are easy to solve on trees!

Infact: Many *NP – Hard* problems can be solved on graphs of bounded treewidth!

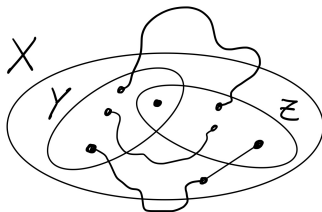
Ex: graph colorings

# Connected Sets

## $k$ -connected sets

A set  $X \subseteq V(G)$  is  **$k$ -connected**, if

- ①  $|X| \geq k$
- ② For all  $Y, Z \subseteq X$  with  $|Y| = |Z| \leq k$  there are  $|Y|$  disjoint  $Y, Z$ -paths



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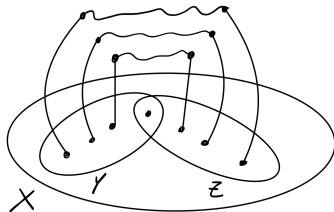
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## Remark

*Horizontal Paths in a  $k \times k$  grid are externally  $k - 1$ -connected*

# Treewidth and Connected Sets

Lemma (Diestel, Gorbunov, Jensen, Thomassen 99')

Let  $G$  be a graph and  $k \in \mathbb{N}_{>0}$ .

- 1 If  $tw(G) < k$  then  $G$  contains no  $(k + 1)$ -connected set of size  $\geq 3k$ .
- 2 If  $h \geq k$  and  $G$  contains no externally  $k$ -connected set of size  $h$  then  $tw(G) < h + k - 1$ .

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Proof 2)

Add picture of current state







## Treewidth and Connected Sets

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### Proof (Continued)

- Claim:  $|X| = h$

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- Claim:  $|X| = h$
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- Let  $v \in C$ .
- Extend tree decomposition  $(\mathcal{B}, T)$  of  $G[U]$ :
  - Add  $t'$  to  $T$  with  $B_{t'} = X \cup \{v\}$ .
  - $(U \cup \{v\}) \cap N_G(V(C - v)) = X \cup \{v\}$

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- Contradicting maximality of  $U$ .









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- Remains to show:  $|(X \cup S) \cap N_G(V(C'))| \leq h$
- $X_Y := (X \setminus Z) \cup S$  and  $X_Z := (X \setminus Y) \cup S$ .

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- $X_Y := (X \setminus Z) \cup S$  and  $X_Z := (X \setminus Y) \cup S$ .
- Claim:  $N_G(V(C')) \subseteq X_Y$  or  $N_G(V(C')) \subseteq X_Z$ .

Add picture proof of claim



# References